

Unit 1: Data Warehouse

Assignment – I / Unit 1 (MCQs)

- 1) OLTP stands for -
 - A. On-Line Transactional Processing
 - B. On Link Transactional Processing
 - C. On-Line Transnational Process
 - D. On-Line Transactional Program
- 2) Data warehouse is -
 - A. The actual discovery phase of a knowledge discovery process
 - B. The stage of selecting the right data for a KDD process
 - C. A subject-oriented integrated time-variant non-volatile collection of data in support of management
 - D. None of these
- 3) A data warehouse is which of the following?
 - A. Can be updated by end-users.
 - B. Contains numerous naming conventions and formats.
 - C. Organized around important subject areas.
 - D. Contains only current data.
- 4) _____ is a subject-oriented, integrated, time-variant, non-volatile collection of data in support of management decisions.
 - A. Data Mining.
 - B. Data Warehousing.
 - C. Web Mining.
 - D. Text Mining.
- 5) The data Warehouse is _____.
 - A. read only.
 - B. write only.
 - C. read write only.
 - D. none.
- 6) Expansion for DSS in DW is _____.
 - A. Decision Support system.
 - B. Decision Single System.
 - C. Data Storable System.
 - D. Data Support System.
- 7) The important aspect of the data warehouse is that data found within the warehouse is _____.
 - A. subject-oriented.
 - B. time-variant.
 - C. integrated.
 - D. All of the above.
- 8) The time horizon in Data warehouse is usually _____.
 - A. 1-2 years.
 - B. 3-4years.
 - C. 5-6 years.
 - D. 5-10 years.
- 9) The data is stored, retrieved & updated in _____.
 - A. OLAP.
 - B. OLTP.
 - C. SMTP.
 - D. FTP.
- 10) _____ describes the data contained in the data warehouse.
 - A. Relational data.
 - B. Operational data.
 - C. Metadata.
 - D. Informational data.

- 11) _____ databases are owned by particular departments or business groups.
A. Informational.
B. Operational.
C. Both informational and operational.
D. Flat.
- 12) The star schema is composed of _____ fact table.
A. one.
B. two.
C. three.
D. four.
- 13) The time horizon in operational environment is _____.
A. 30-60 days.
B. 60-90 days.
C. 90-120 days.
D. 120-150 days.
- 14) The key used in operational environment may not have an element of _____.
A. time.
B. cost.
C. frequency.
D. quality.
- 15) Data can be updated in _____ environment.
A. data warehouse.
B. data mining.
C. operational.
D. informational.
- 16) Record cannot be updated in _____.
A. OLTP
B. files
C. RDBMS
D. data warehouse
- 17) The type of relationship in star schema is _____.
A. many-to-many.
B. one-to-one.
C. one-to-many.
D. many-to-one.
- 18) Fact tables are _____.
A. completely demoralized.
B. partially demoralized.
C. completely normalized.
D. partially normalized.
- 19) Data warehouse architecture is based on _____.
A. DBMS.
B. RDBMS.
C. Sybase.
D. SQL Server.
- 20) _____ is data about data.
A. Metadata.
B. Microdata.
C. Minidata.
D. Multidata.

Subjective Questions

1. Define a Data Warehouse and explain its key features with examples.
2. Differentiate between Operational Databases and Data Warehouses.
3. Describe the various Data Warehouse Models (Enterprise Warehouse, Data Mart, Virtual Warehouse) with diagrams.
4. What are Concept Hierarchies? How are they useful in Data Warehousing?
5. Explain the OLAP operations with examples for each (Roll-up, Drill-down, Slice, Dice, Pivot).
6. Discuss the steps involved in designing and implementing a Data Warehouse.
7. Define Fact and Dimension.
8. What is Star Schema?
9. What is a Galaxy Schema (Fact Constellation)? When is it used?
10. Write the comparison of common Data Warehouse schemas - Star Schema, Snowflake and Galaxy Schema.